INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year

APPLIED DATA STRUCTURES & ALGORITHMS LAB (BCS553)

LAB 10

Assignment 10: Parent and Child Classes

Assignment aim: Use inheritance so child classes can access parent members.

Experiment No. 10.1 — Inherit sound() from Animal

1. Experiment Title: Inherit sound() from Animal

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: Inheritance uses extends to derive a child class from a parent class; super can invoke a parent constructor.

5. Algorithm / Procedure:
Define sound() in Animal; extend Animal with Dog; create Dog and call its inherited method.
Then compile and execute the Java program.

6. Program:

```
class Animal {
    void sound() { System.out.println("Some sound"); }
}
class Dog extends Animal { }
public class Task_01_Inherit_sound_from_Animal {
    public static void main(String[] args) {
        Dog d = new Dog();
        d.sound();
    }
}
```

7. Input: Fixed input values are declared in the program: Animal/Person/Student/Teacher objects and names specified in the task.

8. Output:
Some sound

Time / Space Complexity: O(1) time; O(1) auxiliary space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- What is inheritance?
- What does extends mean?
- Can a child class use accessible parent methods?

Experiment No. 10.2 — Call parent constructor with super(name)

1. Experiment Title: Call parent constructor with super(name)

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: Inheritance uses extends to derive a child class from a parent class; super can invoke a parent constructor.

5. Algorithm / Procedure:
Person stores a name through its constructor; Student extends Person and calls super(name); create and display a Student.
Then compile and execute the Java program.

6. Program:

```
class Person {
    String name;
    Person(String name) { this.name = name; }
}
class Student extends Person {
    Student(String name) { super(name); }
}
public class Task_02_Call_parent_constructor_with_supername{
    public static void main(String[] args) {
        Student s = new Student("Neha");
        System.out.println("Name: " + s.name);
    }
}
```

7. Input: Fixed input values are declared in the program: Animal/Person/Student/Teacher objects and names specified in the task.

8. Output:
Name: Neha

Time / Space Complexity: O(1) time; O(1) auxiliary space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- What does super(name) call?
- Why must a superclass constructor be called?
- What is a child class?

Experiment No. 10.3 — Use inherited greet() in Teacher

1. Experiment Title: Use inherited greet() in Teacher

2. Aim: Implement the task specified in the BCS553 lab assignment.

3. Software / Tools Required: Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

4. Theory: Inheritance uses extends to derive a child class from a parent class; super can invoke a parent constructor.

5. Algorithm / Procedure:
Define name and greet() in Person; create Teacher; set its inherited name field; call greet().
Then compile and execute the Java program.

6. Program:

```
class Person {
    String name;
    void greet() { System.out.println("Hello, " + name); }
}
class Teacher extends Person { }
public class Task_03_Use_inherited_greet_in_Teacher{
    public static void main(String[] args) {
        Teacher t = new Teacher();
        t.name = "Kiran";
        t.greet();
    }
}
```

7. Input: Fixed input values are declared in the program: Animal/Person/Student/Teacher objects and names specified in the task.

8. Output:
Hello, Kiran

Time / Space Complexity: O(1) time; O(1) auxiliary space.

9. Result: The program implements the specified task and produces the expected output for the given test data.

10. Viva-Voce Questions:

- What is an inherited member?
- What is the role of extends?
- How does greet() access name?